Zhuoli Ouyang

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Education

Southern University of Science and Technology — **Shenzhen, China**
Undergraduate, Information Engineering — 2023 to 2027 (expected)
GPA 3.88/4.00 **Weighted Average Score:** 92.89 **Rank** 6/48 **TOEFL** Total 5.0

- **Fields Elite Class (Honors)**, Department of Mathematics — 2023 to 2024
- **Honors Class**, Department of Electronic & Electrical Engineering — 2024 to Present

Research Interests

With a **solid mathematical foundation**, **interdisciplinary knowledge of computational science**, and **hands-on experience in large-scale model training**, I aim to build **efficient and reliable AI systems**. My recent research focuses on understanding the learning dynamics of large models, particularly in LLM pre-training and post-training, and turning these insights into principled algorithms. Building on this, I plan to extend my research to agentic systems that explore their environments autonomously and improve themselves through interaction.

Honors & Awards

2026 ALT 2026 Best Student Paper Award
2026 DATE 2026 Oral Paper

Selected Publications (*Equal contribution)

ENGINE Endogenous Variational MultiScale Optimization for Zeroth-Order LLM Fine-Tuning
Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu — *NeurIPS 2026*

RMNP Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization
Shenyang Deng*, **Zhuoli Ouyang***, Tianyu Pang, Zihang Liu, Ruochen Jin, Shuhua Yu, Yaoqing Yang — *ICML 2026*

Depth, Not Data An Analysis of Hessian Spectral Bifurcation
Shenyang Deng*, Boyao Liao*, **Zhuoli Ouyang***, Tianyu Pang, Yaoqing Yang — *ISIT 2026*

Suspicious Alignment of SGD A Fine-Grained Step Size Condition Analysis
Shenyang Deng, Boyao Liao, **Zhuoli Ouyang**, Tianyu Pang, Minhak Song, Yaoqing Yang — *ALT 2026 (Best Student Paper Award)*

Understanding Generalization from Embedding Dimension and Distributional Convergence
Junjie Yu*, **Zhuoli Ouyang***, Haotian Deng*, Chen Wei, Wenxiao Ma, Jianyu Zhang, Zihan Deng, Quanying Liu — *ICML 2026*

OrthVMS Orthogonal Variational Multiscale Correction for Full-Rank Updates with Low-Rank Optimizer States
Zhuoli Ouyang, Yujun He, Hongbin Zhong, Han Wu, Haoning Jiang, Zhihai He — *ICLR 2027 (Under Review)*

NOVA Nullspace Dynamics in Neural Basis Solvers for Moving Boundary PDEs, Transport, Rank Transitions, and Reprojection
Zhuoli Ouyang, Changxi Chi, Siyuan Li, Tailin Wu — *ICLR 2027 (Under Review)*

Full publication list on my homepage

Research Experience

Westlake University — **Hangzhou, China**
Visiting Student, Advisor Prof. Tailin Wu — Feb 2026 to Aug 2026

Dartmouth College — **Hanover, NH**
Visiting Student, Advisor Prof. Yaoqing Yang — Mar 2025 to Feb 2026

Southern University of Science and Technology — **Shenzhen, China**
Undergraduate Researcher, Advisor Prof. Zhihai He — 2024 to Present

Southern University of Science and Technology — **Shenzhen, China**
Undergraduate Researcher, Advisor Prof. Quanying Liu — Summer 2025

Selected Research Projects

ENGINE Zeroth-order LLM fine-tuning *(First Author, Project Leader)* *NeurIPS 2026*

Advisor Prof. Tailin Wu

My Contributions:

- Proposed ENGINE, coupling LoRA and full parameter spaces as coarse and fine scales via variational multiscale theory.
- Derived a bias-variance analysis showing how this coupling reduces both gradient variance and curvature bias.
- Implemented ENGINE and baselines, achieving SOTA performance among zeroth-order methods.
- Wrote the complete manuscript and led the rebuttal through to acceptance.

RMNP Efficient matrix-based optimizer *(Co-first Author)* *ICML 2026*

Advisor Prof. Yaoqing Yang

My Contributions:

- Replaced Muon's Newton-Schulz iterations with row-wise momentum normalization at a fraction of the cost.
- Built the complete codebase and ran all benchmarks and ablations, achieving competitive performance with Muon.
- Organized the experimental results and wrote the experimental sections and appendix.
- Conducted the rebuttal experiments and drafted the responses to reviewers.

Depth, Not Data Hessian spectral bifurcation *(Co-first Author)* *ISIT 2026*

Advisor Prof. Yaoqing Yang

My Contributions:

- Investigated whether Hessian spectral bifurcation stems from data covariance, as widely assumed, or from network depth.
- Conducted extensive experiments on whitened data across depths, showing that depth alone induces the bifurcation.
- Verified the main theorem against the experiments and caught a derivation error before submission.

Suspicious Alignment of SGD Step-size analysis *(Co-author)* *ALT 2026 (Best Student Paper Award)*

Advisor Prof. Yaoqing Yang

My Contributions:

- Studied why SGD updates along steep Hessian directions can fail to reduce the loss despite gradient alignment.
- Conducted experiments to validate the derived step-size conditions that characterize this behavior.

Understanding Generalization Embedding dimension and generalization *(Co-first Author)* *ICML 2026*

Advisor Prof. Quanying Liu

My Contributions:

- Derived a generalization bound linking embedding dimension and downstream sensitivity to population risk.
- Explained why compressing representations does not always improve generalization.
- Led the literature research and guided experiments across architectures and datasets.

OrthVMS Memory-efficient LLM optimizer *(First Author, Project Leader)* *ICLR 2027 (Under Review)*

Advisor Prof. Zhihai He

My Contributions:

- Proposed OrthVMS, applying VMS theory from computational mechanics to recover discarded gradient information.
- Derived the convergence analysis and designed adaptive updates for the projected space and its orthogonal complement.
- Implemented OrthVMS and baselines, achieving SOTA performance among evaluated memory-efficient optimizers.
- Wrote the complete manuscript.

NOVA Neural PDE solver for moving boundaries *(First Author, Project Leader)* *ICLR 2027 (Under Review)*

Advisor Prof. Tailin Wu

My Contributions:

- Designed NOVA, a training-free solver that enforces moving boundary constraints through nullspace decomposition.
- Derived a nullspace transport equation that removes the need for a full SVD at every time step.
- Evaluated NOVA on eight multiphysics benchmarks, substantially reducing boundary errors with competitive accuracy.
- Implemented NOVA and all baselines, and then wrote the full paper.

Skills

Programming	Python, C/C++, CUDA, MATLAB, LaTeX
Tools & Frameworks	PyTorch, PETSc, Linux, Git, Docker, OpenFOAM